

Formal Feature Attribution as a Causal Calculator for Drug-Drug Interaction Risk in Geriatric Polypharmacy: Evidence from a State-Level All-Payer Claims Database

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Background: Polypharmacy (≥ 5 concurrent medications) affects more than 40% of adults aged ≥ 65 years and drives adverse drug events (ADEs) responsible for an estimated 100,000 hospitalizations annually. Pairwise DDI databases flag co-prescribed pairs in isolation, missing synergistic multi-drug effects where each constituent pair appears benign.

Methods: Using Virginia's All-Payer Claims Database (APCD; 6,929,576 patients; 2016–2019), we constructed a polypharmacy cohort ($n = 1,182$ geriatric cases; 65–114 years; 5:1 matched controls). A 30-day causality window isolated proximal drug exposures. A per-event-density ensemble (CatBoost/XGBoost/XGBoost-RF; Optuna-tuned; training: 2016–2018; holdout: 2019) fed Formal Feature Attribution (FFA) extended to multi-feature interaction testing of drug pairs and triplets, with Intervention Rate (IR) scores quantifying expected risk reduction per deprescribing action. FP-Growth network visualization contextualized causal findings within drug co-occurrence topology.

Results: All 8 age bands achieved PR-AUC ≥ 0.908 (peak: 0.997 at 75–84 band). FFA identified 115 synergistic drug pairs and 5,021 high-risk triplets invisible to pairwise DDI databases. Levofloxacin acted as a pharmacological hub (CYP1A2 inhibition), appearing in 4 of the top 5 synergistic pairs: acetaminophen + levofloxacin (IE = 16.3) and levofloxacin + lorazepam (IE = 11.9). Top IR-ranked deprescribing targets: simvastatin (IR 7.0×10^{-4}), furosemide (IR 2.0×10^{-4}), alprazolam (IR 1.0×10^{-4}). A U-shaped Z-code monitoring effect identified a fragmented-care phenotype (Q4; OR ≈ 0.94 vs. unmonitored) distinct from protectively monitored patients (Q2; OR = 0.25).

Conclusions: FFA multi-feature interaction analysis constitutes a causal calculator for polypharmacy ADE risk, detecting synergistic combinations invisible to pairwise DDI databases and yielding IR-ranked deprescribing priorities from standard claims data.

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